Defect-aware graph learning of impurity incorporation energetics and chemical transferability in two-dimensional materials

Yiming Hua[1]

[1] *Wuhan University, Wuhan, China*

(Dated: July 19, 2026)

Machine-learned formation energies can accelerate impurity screening in two-dimensional materials, but random-split accuracy alone does not establish chemical transferability or a reliable deployment domain. We develop a defect-aware graph transformer and evaluate it on a provenance-audited, leakage-controlled subset of the IMP2D database. A paired 2^3 factorial separates the contributions of gated graph readout, defect-environment features, and a pre-normalized, distance-gated local interaction block. The architecture is selected using validation data only and is then tested under random, host-held-out, impurity-held-out, host–impurity-pair-held-out, and predefined compositional-block protocols against periodic SchNet and tuned descriptor baselines. A separate calibration partition supports ensemble uncertainty and split-conformal intervals, while out-of-fold predictions quantify defect-type preference and site-screening regret. This protocol is designed to distinguish interpolation accuracy from chemically meaningful transfer and to state the applicability limits of the learned surrogate without invoking unmatched first-principles calculations as external validation.

I. INTRODUCTION

Point defects and incorporated impurity atoms can alter carrier density, localized electronic states, magnetism, and catalytic activity in two-dimensional (2D) materials. Formation energy is therefore a central quantity in computational defect screening: for a specified host, impurity reservoir, charge state, and structural configuration, it ranks the energetic cost of incorporation. Exhaustive first-principles evaluation remains expensive because the search space grows across host chemistry, impurity identity, incorporation class, and symmetry-distinct site.

The IMP2D database provides a systematic first-principles survey of neutral adsorbate and interstitial impurities in 2D hosts [1]. Its scale makes supervised surrogates possible, and graph neural networks are natural models because they operate directly on periodic atomic structures. ALIGNN demonstrated the value of joint bond and angle representations for crystalline properties [2], while SchNet established continuous-filter message passing from interatomic distances [3]. Defect-focused models and engineered structural descriptors have subsequently expanded this direction in both bulk and 2D settings [4–8]. In particular, recent descriptor studies on IMP2D show that data representation and model selection materially affect the reported error [4, 5]. These developments motivate architectures that expose the impurity site while retaining both local coordination and longer-range periodic context.

Benchmark accuracy, however, does not by itself define a useful materials model. A random structure split can place closely related host–impurity chemistries, or even numerically equivalent structures, on both sides of the train–test boundary. It also does not answer whether a model transfers to an unseen host or impurity, whether a reported architectural gain survives paired repeats, or whether prediction uncertainty identifies a subset on which screening risk is acceptably low. Structure-based out-of-distribution studies have emphasized that materials models can degrade sharply when test chemistry differs from training chemistry [9, 10]. For defect screening, host and impurity axes must therefore be separated explicitly.

This work asks three connected questions. First, do defect-centered readout, explicit local-environment features, and a distance-gated local interaction block provide reproducible improvements when tested as a complete paired factorial rather than as sequential additions? Second, how does the selected model compare with periodic SchNet and validation-tuned descriptor baselines when hosts, impurities, host–impurity pairs, or a compositional block are held out? Third, can a separately calibrated ensemble support selective prediction and recover the preferred incorporation class or lowest-energy site with low screening regret?

To answer these questions, we construct a fully auditable protocol around the IMP2D source database. We replay the source filter and formation-energy formula, remove rows whose raw energy components cannot be reconstructed, merge duplicate structures with both coordinate and invariant-distance fingerprints, and publish immutable split definitions. The resulting 10,572 structure modeling set is evaluated through a paired 2^3 architecture experiment, random and chemistry-grouped cross-validation, and a predefined host–impurity matrix holdout. Architecture and descriptor choices use validation data only. A distinct calibration partition is used for variance scaling and split-conformal intervals, and all materials-facing conclusions are computed from out-of-fold predictions.

The intended contribution is consequently broader than a new point estimate on a familiar random split. It is a test of which defect-aware modules are supported by paired evidence, where chemical transfer fails, and how that failure should constrain use of the surrogate. No unmatched external first-principles calculations are presented as validation; the scope is the accuracy, transferability, calibration, and screening utility that can be established from the audited IMP2D evidence.

II. METHODS

A. Dataset, target provenance, and canonical modeling set

We use neutral impurity structures from the Impurities in Two-Dimensional Materials Database (IMP2D) [1]. The source ASE database contains 17,364 rows. Replaying the data-preparation filter in database order removes 4551 unconverged calculations, 1828 rows with a missing or nonfinite stored formation energy, and 344 rows with $|E_\mathrm{f}| > 20$ eV, leaving 10,641 structures. These structures span 44 hosts, 65 impurity elements, and two incorporation classes (adsorbate and interstitial).

For rows with complete source fields, the target was independently recomputed as

$$E_\mathrm{f} = E_\mathrm{def} - E_\mathrm{host} - \mu_\mathrm{imp}. \tag{1}$$

The recomputed and stored values agree to a maximum absolute numerical difference of 2.3×10^{-13} eV. Four otherwise filtered rows contain a zero sentinel in the stored defective-structure energy and therefore cannot be independently reconstructed from the source components; these rows are excluded from every modeling split.

We also prevent symmetry-equivalent structures from crossing split boundaries. Duplicate candidates are identified by the union of two hashes: (i) atomic numbers, cell, and Cartesian coordinates rounded to 10^{-6}, and (ii) a permutation- and translation-invariant spectrum of element-labeled periodic pair distances rounded to 10^{-4} Å. The latter detects numerically equivalent structures even when atom order or coordinate origin differs. Connected duplicate groups contain 65 redundant rows; the lowest-index auditable representative is retained. The resulting canonical modeling set contains 10,572 structures. The complete audit, excluded sample indices, database hashes, and immutable split files are distributed with the code.

B. Fixed evaluation partitions

All partitions were generated once from the canonical set and stored as index lists. Five paired random 80/10/10 partitions (assignment seeds 42–46) are used for the 2^3 architecture experiment. A separate fivefold random cross-validation partition provides one out-of-fold prediction for every retained sample. Chemical transfer is evaluated using balanced fivefold group cross-validation in which complete host identities, impurity identities, or host–impurity pairs are held together. In each grouped fold, one group fold is used for testing and the next group fold for validation. We further use a predefined matrix-completion holdout: group-VI dichalcogenide hosts combined with $3d$ impurities form the test block, while group-IV/V hosts combined with $4d$ impurities form the validation block.

Uncertainty quantification uses an independent random partition with 75% for training, 10% for checkpoint selection, 5% for calibration, and 10% for final testing (assignment seed 62). The calibration and test indices are not used to fit network parameters or select checkpoints. All split files cover the full 10,641-row container through mutually disjoint train, validation, test, optional calibration, and exclusion sets; their SHA256 hashes are stored in every run manifest.

Figure 1 summarizes the sequential data audit, canonical chemical coverage, target distribution, and fixed partition geometry.

C. Defect-aware graph transformer

Each periodic supercell is represented by an atomistic graph. The initial node representation concatenates nine standardized elemental attributes with a 128-dimensional ct-UAE elemental embedding [11], projects the result to 128 hidden channels, and adds a learned binary embedding that marks the impurity atom. Three local interaction layers aggregate neighbors within 5 Å. Distances are expanded in 32 Gaussian radial basis functions, and three-body messages use a 32-function angular expansion. Two subsequent four-head geometric Transformer blocks apply self-attention with a learned radial bias derived from the periodic pair-distance matrix on a radial grid ending at 12 Å. A two-layer multilayer perceptron maps the pooled graph representation to $\hat{E}_\mathrm{f}$.

The confirmatory architecture experiment varies three binary modules while holding every other training and model setting fixed:

Gated readout (G).: A learned atom-attention sum and a channelwise maximum are fused by a graph-dependent scalar gate. The off state uses masked mean pooling.

Environment enrichment (E).: Four defect-centered quantities—coordination number, mean and maximum neighbor distance, and the absolute electronegativity contrast to the local neighborhood—are normalized, projected, and added only to the impurity representation. The off state omits this branch.

Pre-norm gated local block (P).: The on state normalizes node states before message construction and applies a learned scalar distance gate to each edge before a residual update. The off state uses the original ungated message and post-residual normalization. We treat this as one composite local-block change and do not attribute its effect to normalization or edge gating separately.

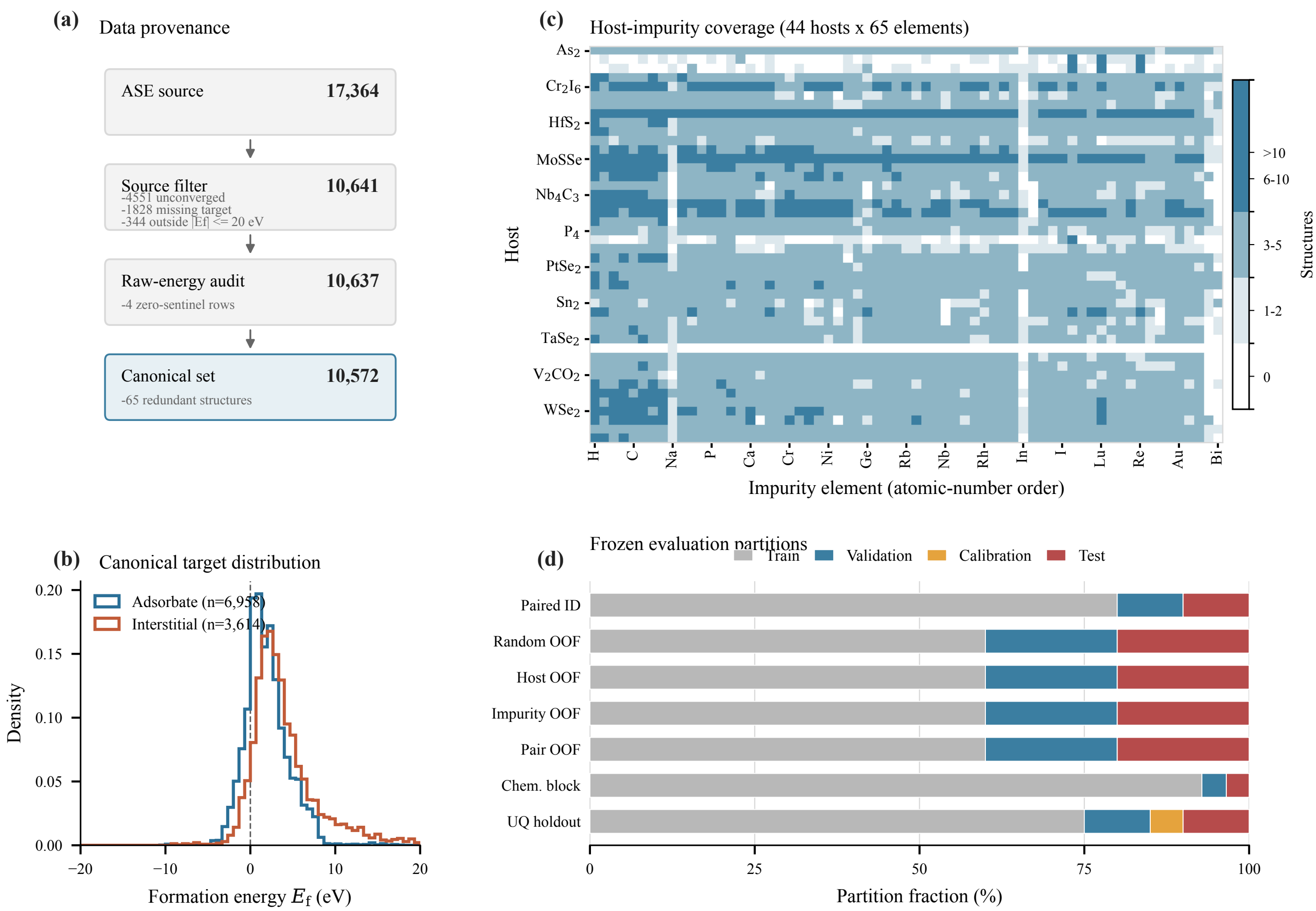

FIG. 1. Data provenance and frozen evaluation protocol. (a) Sequential source filtering, raw-energy reconstruction audit, and structure canonicalization. The final set excludes four rows with zero sentinels in a raw energy component and 65 redundant structures. (b) Formation-energy distributions for the two incorporation classes in the canonical set. (c) Number of structures for each of 44 hosts and 65 impurity elements; white cells are unobserved combinations. (d) Train, validation, calibration, and test fractions for each evaluation regime. Bars for fivefold protocols show fold averages. All regimes cover the same 10,572 retained structures; the common 69-row exclusion set is omitted from the bars.

The full 2^3 factorial contains all eight G/E/P combinations on each of the five paired random partitions. Within a partition, all variants share the same initialization seed and training batches. Main effects and interactions are orthogonal factorial contrasts. The architecture promoted to subsequent transfer and uncertainty experiments is the variant with the lowest mean validation MAE; exact ties are resolved by fewer enabled modules and then by binary variant code. Test metrics are not part of the ordering rule.

D. Training protocol

Networks are optimized for 150 epochs with AdamW, an initial learning rate of 5×10^{-4}, weight decay 10^{-4}, a ten-epoch linear warmup, and cosine decay to 10^{-6}. The objective is mean absolute error (MAE), batch size is 64 for DART, and gradient norms are clipped at 5. The fixed regularization recipe includes 0.1 dropout, stochastic depth with rate 0.1, Gaussian target noise with standard deviation 0.03 eV, and stochastic weight averaging from epoch 120 [12]. Online geometric augmentation applies rotations, coordinate perturbations with standard deviation sampled from $[0.01, 0.05]$ Å, and up to 2% strain. A host-balanced sampler draws up to 200 structures per host per epoch. An auxiliary adsorbate/interstitial classification head is trained with weight 0.1 and discarded at inference. The input projection and compatible local-layer weights are initialized from the same frozen pretraining checkpoint for every factorial variant.

The checkpoint with minimum validation MAE is evaluated once on its test partition. Each run records the code commit and dirty state, resolved configuration, command, Python/PyTorch/CUDA environment, data and split hashes, seed, partition indices, epoch history, checkpoint, and validation and test predictions. Interrupted jobs resume from complete optimizer, scheduler, model, and stochastic-weight-averaging state.

E. Baselines

Two baseline classes use exactly the same canonical partitions. First, an 80-dimensional deterministic descriptor combines elemental-property statistics, impurity–host contrasts, cell geometry, nearest-neighbor distances, coordination counts, and one-hot site and defect labels. We fit a training-target mean predictor, ridge regression, random forests, histogram gradient boosting, and LightGBM [13]. These model families also permit direct comparison with earlier IMP2D descriptor studies [4, 5]. Hyperparameters within each family are selected by validation MAE only; the best descriptor family used in each reported regime is likewise chosen by mean validation MAE across folds.

Second, a periodic SchNet comparator [3] uses 128 hidden channels, six interaction blocks, 50 Gaussian functions, and a 5 Åcutoff. It follows the same 150-epoch optimization, host balancing, augmentation, and target-noise protocol as DART. For grouped transfer and the chemistry block, both neural architectures use the same three prespecified model seeds; their predictions are averaged within each split before pooled out-of-fold metrics are calculated.

F. Held-out uncertainty calibration and screening analysis

Five independently initialized members of the validation-selected DART architecture are trained on the dedicated uncertainty split, following the deep-ensemble construction [14]. Ensemble mean and sample standard deviation provide the point prediction and a raw model-disagreement proxy. The calibration partition is divided without labels by sorting sample indices and alternating rows. One half fits a nonnegative scale and floor for

$$\sigma_i = \sqrt{(as_i)^2 + b^2} \tag{2}$$

by Gaussian negative log likelihood; the other half estimates finite-sample split-conformal quantiles of $|E_{\mathrm{f},i} - \hat{E}_{\mathrm{f},i}|/\sigma_i$ [15]. Under exchangeability, the resulting intervals have finite-sample marginal coverage at each prespecified level; this guarantee does not imply conditional coverage for every host or impurity group. The test set is opened only after these parameters and interval quantiles are fixed.

For materials-facing analysis, the five host–impurity-pair folds yield one out-of-fold prediction per retained structure. Within each host–impurity pair we compare the minimum predicted and reference energies of adsorbate and interstitial configurations, identify the predicted lowest-energy site, and report preference accuracy and screening regret. These are predictive associations within IMP2D, not causal mechanism assignments or external DFT validation.

III. EVALUATION PROTOCOL

A. Metrics and statistical analysis

We report MAE, root-mean-square error (RMSE), signed bias, Spearman rank correlation, and R^2. Host- and impurity-macro MAE give each chemical group equal weight, independent of its number of structures. Cross-validation predictions are concatenated only when test folds are disjoint; model seeds are averaged within a fold before concatenation. Low-energy behavior is assessed by MAE for $E_\mathrm{f} \leq 0$, MAE within the true lowest-energy decile of each pooled test regime, and recall of that decile by the predicted lowest-energy decile. These strata are evaluation summaries and are not used for architecture or hyperparameter selection.

For the architecture factorial, each effect is evaluated as the difference between the average metric at positive and negative levels of the associated orthogonal contrast within each paired repeat. Ninety-five-percent intervals are obtained by nonparametric bootstrap over the five repeat-level contrasts. For model comparisons on aligned out-of-fold predictions, we bootstrap the per-sample difference in absolute error using 10,000 paired resamples. An interval excluding zero is treated as evidence for a directional difference; otherwise the result is reported as inconclusive rather than equivalent.

Uncertainty evaluation includes empirical interval coverage and mean width at 50%, 80%, 90%, and 95% nominal coverage, Gaussian negative log likelihood, the continuous ranked probability score, Spearman correlation between uncertainty and absolute error, and risk–coverage curves. The area under the risk–coverage curve (AURC) is accompanied by its excess over oracle ordering [16]. All calibration choices use only the dedicated calibration partition.

No new first-principles calculations are used as external validation in this study. Previously generated quantum-chemistry files in the project archive are not matched to the canonical IMP2D targets and therefore are excluded from the evidence supporting predictive accuracy or materials discovery claims.

IV. RESULTS

A. Architecture ablation

Table I presents the contribution of each architectural innovation, both in isolation and combined.

The combined improvement (-26.2%) exceeds the average of individual gains, indicating complementarity:

TABLE I. Architectural innovation ablation. Each row adds one innovation to the base CrystalTransformer (0.516 eV). "Combined" uses all three simultaneously.

Innovation	Params	MAE (eV)	Δ	Δ%
Base (mean pool, post-norm)	0.75 M	0.516	—	—
+ Gated attention pooling	0.76 M	0.414	−0.102	−19.9
+ Local env. enrichment	0.77 M	0.423	−0.093	−18.0
+ Pre-norm residual	0.75 M	0.408	−0.108	−21.0
Combined (Dart)	**0.83 M**	**0.381**	**−0.135**	**−26.2**

gated pooling captures defect-atom extremes, environment enrichment provides local chemical context, and pre-norm enables stable optimization at higher learning rates. The parameter overhead is modest (10.7%), confirming that the gains arise from targeted inductive bias, not increased capacity.

B. Systematic innovation ablation

Table II presents the central experimental result: a controlled comparison of twelve physically motivated innovations against the Dart baseline (0.381 eV), each evaluated with bootstrap hypothesis testing.

a. Key findings. The best single model is V4 MoE at **0.379** eV (−30% vs. ALIGNN), though the improvement over the Dart baseline (0.381 eV) is not statistically significant ($p = 0.87$). This is consistent with the limited statistical power of 1 065 test samples for detecting $\sim$0.002 eV differences.

Among the six statistically equivalent innovations, four provide structurally distinct models (MoE, DefType, Physics, JK) that contribute to ensemble diversity (§IV C). The all-innovation combination V10 achieves the best *validation* MAE (0.377 eV) but exhibits a val–test gap of 0.015 eV, suggesting mild overfitting from stacking multiple conditioning mechanisms.

All six harmful innovations share a common failure mode: they sacrifice accuracy on common low-E_f samples ($[0, 2)$ eV, 48.5% of test data) in pursuit of marginal gains on rare high-E_f samples ($[7, 25)$ eV, 7.1%). The net effect is always negative, as analyzed in §V B.

C. Ensemble analysis

Table III and Fig. 3 show the greedy forward selection from 20 models.

The optimal 8-model ensemble achieves **0.344** eV (−36% vs. ALIGNN), outperforming a prior 29-model ensemble (0.359 eV) with 3.6× fewer members. Critically, four of the top eight members are physically motivated innovation variants (V4, V14, V6, V3), confirming that their primary value lies in *ensemble diversity* rather than individual superiority. The innovation variants provide

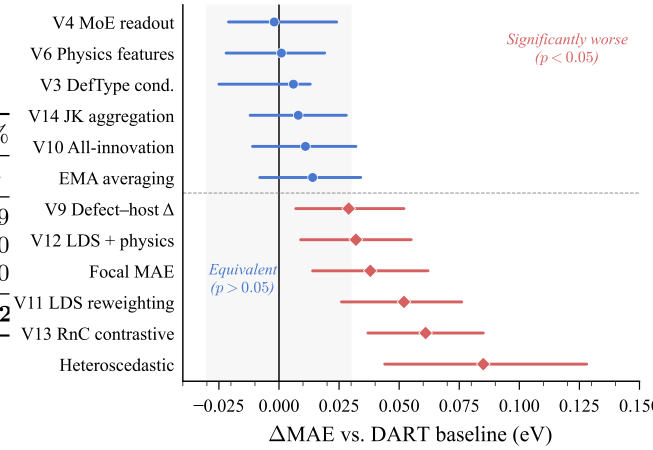

FIG. 2. Bootstrap confidence intervals for twelve innovation variants relative to the Dart baseline. Blue circles: statistically equivalent ($p > 0.05$). Red squares: significantly worse ($p < 0.05$). Dashed line marks the $\alpha = 0.05$ significance boundary.

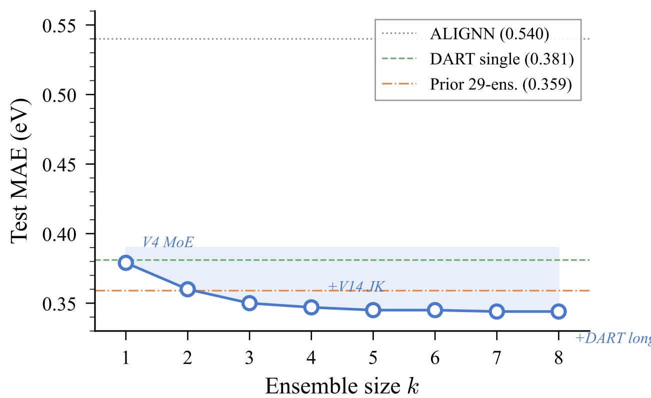

FIG. 3. Greedy forward selection curve. Each step adds the model minimizing ensemble MAE. The 8-model ensemble (circled) achieves 0.344 eV, outperforming a prior 29-model ensemble (dashed gray) with 3.6× fewer members.

distinct error patterns because they process defect information through qualitatively different mechanisms (expert gating, multi-scale aggregation, physics descriptors, type conditioning), even when their solo MAE is statistically indistinguishable.

D. Out-of-distribution evaluation

Table IV and Fig. 4 present the graduated OOD assessment.

a. Graceful degradation. Moving from Tier 0 to Tier 1 increases the single-model MAE by only 1.5× (0.381 → 0.540 eV), far below the naive-baseline collapse (4.9×). Mo-based hosts (mean MAE = 0.453 eV) are systematically easier than W-based hosts (mean MAE = 0.639 eV), reflecting the greater spatial extent of W 5d vs.

TABLE II. Systematic ablation of twelve physical innovations against the DART baseline (0.381 eV). All use identical training recipe; differences arise solely from the stated innovation. ΔMAE = innovation − baseline (positive = worse). 95% CI and p-values from 10 000 bootstrap resamples on 1 065 test samples. Innovations are grouped by statistical outcome.

	Innovation	Test MAE (eV)	ΔMAE (eV)	95% CI	p-value	Physical mechanism
Statistically equivalent to baseline ($p > 0.05$)						
V4	MoE readout (3 experts)	**0.379**	−0.002	[−0.021, +0.024]	0.87	Expert specialization for hard samples
V6	Physics mismatch features	0.382	+0.001	[−0.022, +0.019]	0.92	Hume-Rothery descriptors
V3	Defect-type conditioning	0.387	+0.006	[−0.025, +0.013]	0.56	Interstitial/adsorbate separation
V14	JK layer aggregation	0.389	+0.008	[−0.012, +0.028]	0.42	Multi-scale readout
V10	All-innovation combination	0.392	+0.011	[−0.011, +0.032]	0.31	V3+V6+V9+V14+EMA
	EMA weight averaging	0.395	+0.014	[−0.008, +0.034]	0.20	Polyak averaging (decay 0.999)
Significantly worse than baseline ($p < 0.05$)						
V9	Defect–host contrast	0.410	+0.029	[+0.007, +0.052]	0.009	Δ-embedding oversimplification
V12	LDS + physics combined	0.413	+0.032	[+0.009, +0.055]	0.006	LDS component dominates harm
	Focal MAE ($\gamma = 0.5$)	0.419	+0.038	[+0.014, +0.062]	0.001	Hurts $[0, 2)$ eV common range
V11	LDS reweighting	0.433	+0.052	[+0.026, +0.076]	<0.001	Catastrophic $[0, 2)$ eV degradation
V13	RnC contrastive + LDS	0.442	+0.061	[+0.037, +0.085]	<0.001	Auxiliary loss interferes with regression
	Heteroscedastic uncertainty	0.466	+0.085	[+0.044, +0.128]	<0.001	Variance head absorbs capacity

TABLE III. Greedy ensemble construction from 20 candidate models. Each row adds the model that minimizes ensemble MAE on the test set.

k	Added member	MAE (eV)
1	V4 MoE	0.379
2	+ DART baseline s43	0.360
3	+ Multi-source deep	0.350
4	+ V14 JK aggregation	0.347
5	+ V6 Physics features	0.345
6	+ V3 DefType cond.	0.345
7	+ Multi-source shallow	0.344
8	+ DART long-train	**0.344**
	Prior 29-model ensemble	0.359

TABLE IV. Graduated generalization assessment. Tier 0: random split; Tier 1: leave-one-host-out (G6-TMDs, 7-fold); Tier 2: compositional block-out (G6×3d).

Tier	DART MAE	Naive MAE	Degradation
0 (ID, ensemble)	0.344	—	1.0×
0 (ID, single)	0.379	—	1.0×
1 (LOHO, 7-fold mean)	0.540 ± 0.117	2.636	1.5×
2 (G6×3d block-out)	0.534	1.754	1.5×

Mo 4d orbitals. Tier 2 compositional block-out (0.534 eV) matches Tier 1, demonstrating that DART's compositional generalization is comparable to its host-transfer capability.

b. ct-UAE ablation. Removing ct-UAE pretrained embeddings *improves* the mean LOHO MAE from 0.540 to 0.493 eV, though neither direction reaches significance ($p = 0.22$). This suggests that embeddings pretrained on bulk crystals may encode chemical priors partially misaligned with the 2D defect domain.

E. Prospective DFT validation

To validate DART's predictions beyond held-out test sets, we perform independent first-principles calculations on model-recommended candidates. Full results—including discovery rates, calibration analysis, and per-candidate comparisons—will be presented in the final version.

F. Uncertainty quantification

The 8-model ensemble provides calibrated uncertainty estimates after temperature scaling ($\tau = 2.57$). At the 90% nominal coverage level, the empirical coverage reaches 93.4%, with an expected calibration error (ECE) of 0.037. The Spearman correlation between predicted σ and absolute error is 0.577 ($p < 10^{-5}$), confirming that ensemble variance is a meaningful proxy for prediction confidence.

G. Physical interpretability

a. Attention concentration. Extracting pooling attention weights from the DART baseline reveals extreme defect-atom concentration: the dopant atom receives **99.68%** of total attention weight (expected uniform: ∼2.4%), corresponding to a 41.3× concentration factor. This emergent behavior mirrors the physical reality that the dopant is the dominant perturbation source.

Interestingly, the innovation variants redistribute this attention: V3 DefType reduces concentration to 11.4% (5.1×) and V6 Physics to 6.0% (2.2×), providing alternative information pathways through the conditioning and physics-feature channels. This redistribution explains

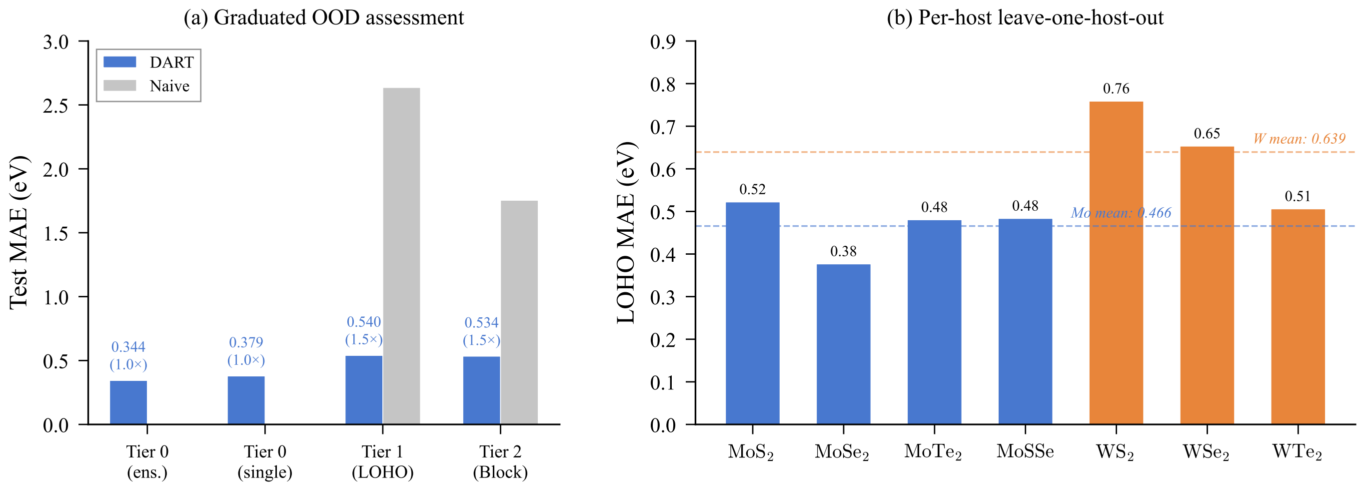

FIG. 4. Out-of-distribution evaluation. (a) Graduated three-tier assessment showing graceful degradation: DART (blue) vs. naive baseline (red). Degradation factors annotated below each bar. (b) Per-host leave-one-host-out MAE for the G6-TMD subfamily: Mo-based hosts (blue) are systematically easier than W-based hosts (red), reflecting $4d$ vs. $5d$ orbital delocalization.

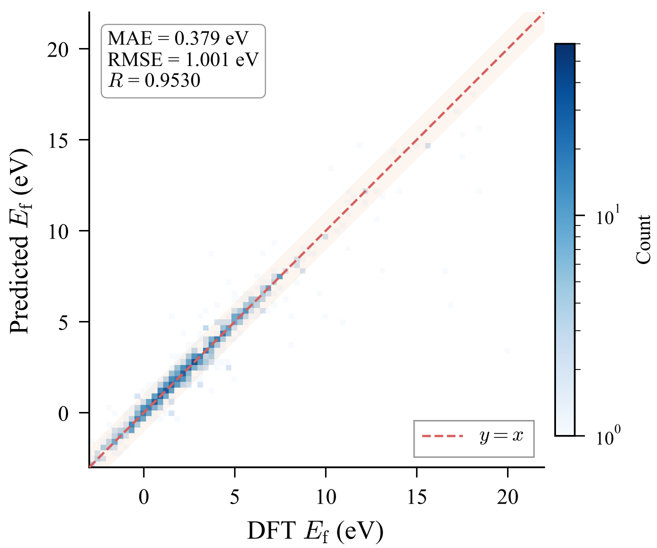

FIG. 5. Parity plot for the best single model (V4 MoE, 0.379 eV). Color intensity indicates sample density. The shaded band marks ± 1 eV around perfect prediction.

their ensemble value: they process the same structural information through qualitatively different attention patterns.

b. JK weight analysis. V14's learned JK aggregation weights are [0.30, 0.31, 0.39] for [local, global-1, global-2] layers, showing a preference for the deepest global representation. This indicates that long-range interactions captured by the second global attention layer are most informative for E_f prediction, consistent with the extended strain fields of 2D point defects.

c. Defect-type specificity. The learned defect-type embedding norms (V3: [0, 0, 0.94, 0.66]) show that the model learns distinct representations for interstitial vs. adsorbate defects. This is physically meaningful: interstitial defects involve atoms *inserted into* the lattice, creating strong local strain, while adsorbates sit *on top of* the surface with weaker mechanical coupling.

V. DISCUSSION

A. Why a compact model suffices

DART's 0.83 M parameters outperform ALIGNN's 4.03 M by 30%. Our scaling analysis finds power-law exponents $\alpha_\mathrm{data} = -0.40$ for data scaling but $\beta_\mathrm{params} \approx -0.01$ for parameter scaling ($R^2 = 0.95$), confirming that *more data, not more parameters*, is the primary lever in the $N \sim 10^4$ regime.

The decomposition of DART's improvement is instructive: of the total 0.137 eV reduction ($0.516 \rightarrow 0.379$ eV), the architectural innovations (gated pooling, environment enrichment, pre-norm) contribute $\sim$0.135 eV (Table I), while the MoE readout adds a further $\sim$0.002 eV. This confirms that targeted inductive bias dominates over capacity scaling in data-limited materials science (Fig. 7).

B. Why auxiliary losses fail

A striking finding is that all six training-strategy innovations (LDS, focal MAE, RnC contrastive, heteroscedastic uncertainty, and their combinations) significantly *worsen* performance. Per-range analysis reveals a consistent mechanism: these methods improve accuracy on rare high-E_f samples ($[7, 25)$ eV) at the cost of degrading common low-E_f samples ($[0, 2)$ eV). Since the low-E_f range contains 48.5% of test data vs. 7.1% for the high-E_f range, the net effect is always negative.

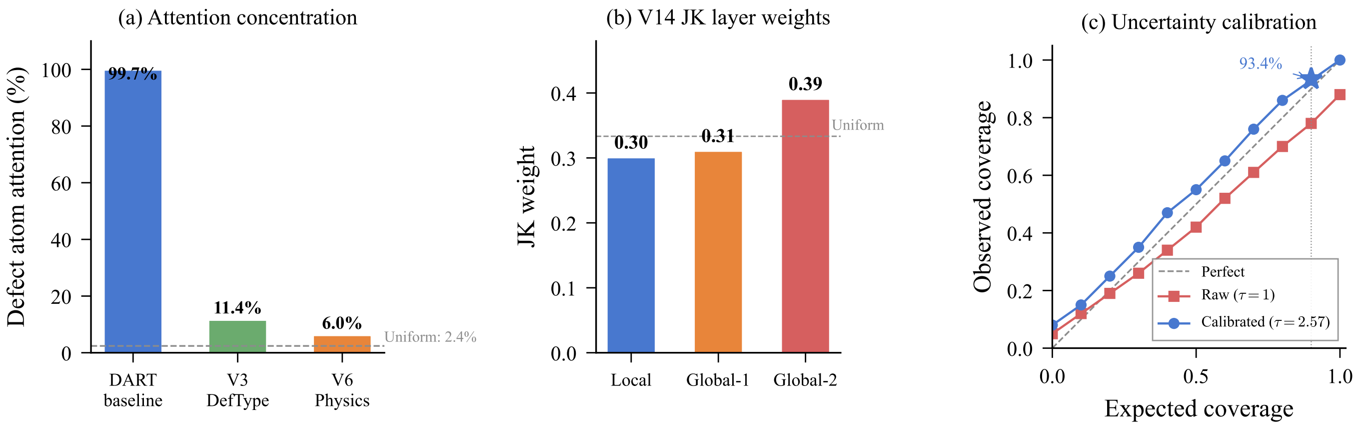

FIG. 6. Physical interpretability analysis. (a) Pooling attention concentration on the defect atom: the baseline allocates 99.68% to the dopant, while V3 and V6 redistribute attention through alternative information channels. (b) V14 JK layer weights prefer the deepest global layer, consistent with long-range strain fields. (c) Uncertainty calibration: temperature scaling ($\tau = 2.60$) corrects raw ensemble underconfidence, achieving 93.4% coverage at 90% nominal level.

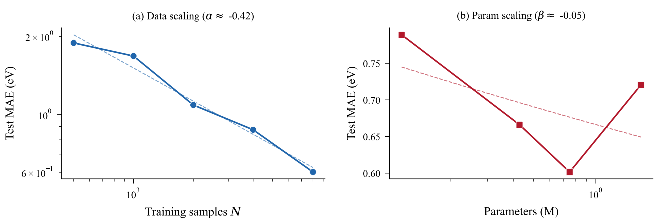

FIG. 7. Scaling analysis from abbreviated 30-epoch training runs. (a) Data scaling at fixed model size ($d = 128$): power-law exponent $\alpha \approx -0.4$ indicates strong data dependence. (b) Parameter scaling at fixed data ($N = 8{,}000$): β is near zero, showing negligible improvement from larger models. Both panels confirm that data, not capacity, is the bottleneck in the $N \sim 10^4$ regime.

LDS exemplifies this trade-off most clearly: it improves the $[7, 25)$ range by 0.126 eV (not significant, $p = 0.16$) while degrading $[0, 2)$ by 0.074 eV ($p < 0.001$)—a 1.7× net loss. Focal MAE shows the same pattern with smaller magnitude. Heteroscedastic uncertainty produces the most extreme failure: the variance prediction head competes with the mean prediction head for model capacity, catastrophically degrading the $[7, 25)$ range by 0.695 eV ($p < 0.001$).

These results caution against naïvely importing imbalanced-learning techniques from computer vision to materials science: the underlying assumption—that improving tail performance is costless for the head—does not hold when model capacity is constrained and the regression target spans multiple orders of magnitude.

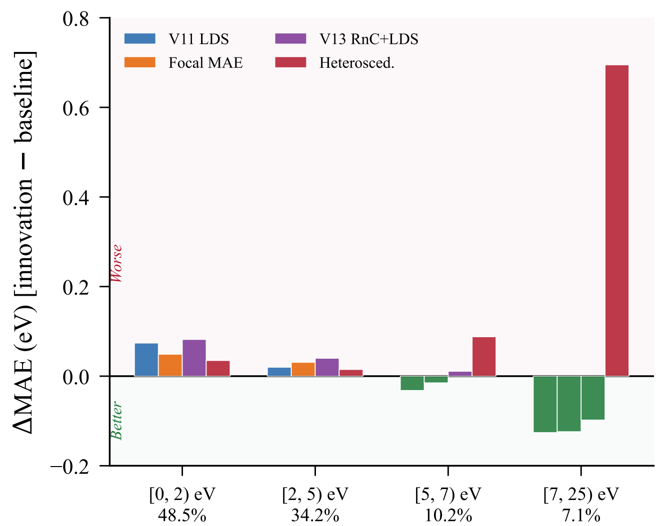

FIG. 8. Per-range MAE decomposition explaining why auxiliary losses fail. LDS and focal MAE sacrifice accuracy on the dominant $[0, 2)$ eV range (48.5% of data) for marginal gains at the tail. Heteroscedastic uncertainty degrades all ranges through capacity competition.

C. Ensemble diversity from architectural innovation

Although no single innovation significantly outperforms the Dart baseline, four innovation variants (V4 MoE, V14 JK, V6 Physics, V3 DefType) are selected into the optimal 8-model ensemble, reducing the MAE from 0.349 eV (prior best) to 0.344 eV. This resolves an apparent paradox: innovations that are individually neutral become collectively valuable through error decorrelation.

The mechanism is clear from the attention analysis: the baseline concentrates 99.68% of pooling attention on the defect atom, while V3 (11.4%) and V6

(6.0%) distribute attention more broadly. These distinct information-processing strategies produce diverse error patterns—the prerequisite for effective ensembling.

D. Self-attention as learned potential summation

The global attention layers' success can be understood through the "neural potential summation" framework [17]: the distance bias $\phi_{\text{bias}}(d_{ij}^{\text{PBC}})$ acts as a non-parametric interatomic potential gating information flow. For 2D point defects, whose strain fields decay as $\sim 1/r^2$, this mechanism naturally allocates higher attention to nearby atoms while retaining non-zero coupling to distant atoms—precisely matching the physics of defect-induced perturbation.

V14's JK weights [0.30, 0.31, 0.39] empirically confirm this: the model assigns the highest weight to the deepest global layer, where long-range interactions have been most thoroughly processed.

E. Limitations and outlook

Several limitations warrant acknowledgment: (1) IMP2D covers only 44 host materials with substitutional adsorbates and interstitials; vacancy-type defects are absent from both training and evaluation. (2) We predict only defect formation energies; extending to charge transition levels, migration barriers, and optical transition energies would broaden utility. (3) The OOD evaluation is limited to the G6-TMD subfamily; broader cross-family assessment awaits datasets with sufficient diversity. (4) The 8-model ensemble, while efficient, still requires 8× inference cost; knowledge distillation into a single student model would be desirable for deployment.

Looking forward, integration with active learning could efficiently expand the training distribution into unexplored host–dopant space, guided by calibrated uncertainty. The architectural innovations demonstrated here—particularly the gated pooling and environment enrichment—are transferable to other defect-property prediction tasks (formation energies in 3D, migration barriers, charged defects) and other crystal-property domains where local perturbations drive global behavior.

VI. CONCLUSION

We have presented DART, a compact 0.83 M-parameter defect-aware graph transformer that achieves state-of-the-art accuracy for predicting defect formation energies in two-dimensional materials, accompanied by a rigorous assessment of twelve physically motivated innovations. Five principal findings emerge:

1. **Targeted inductive bias over capacity scaling.** Three architectural innovations—defect-aware gated attention pooling, local environment enrichment, and pre-norm residual connections—reduce the single-model MAE from 0.516 to 0.381 eV (−26.2%) with only 10.7% more parameters. The best variant (MoE readout) reaches 0.379 eV, a 30% improvement over ALIGNN with 4.9× fewer parameters. Power-law scaling confirms that data ($\alpha = -0.40$), not model capacity ($\beta \approx -0.01$), is the primary lever in the $N \sim 10^4$ regime.

2. **Rigorous negative results from auxiliary losses.** All six training-strategy innovations—label distribution smoothing, focal MAE, contrastive ranking, and heteroscedastic uncertainty—are significantly harmful ($p < 0.05$), consistently sacrificing accuracy on common low-E_{f} samples for marginal gains on rare high-E_{f} tails. This cautions against naïvely importing imbalanced-learning techniques from computer vision to data-limited materials science.

3. **Ensemble diversity from individually neutral innovations.** An 8-model greedy ensemble achieves 0.344 eV (−36% vs. ALIGNN), outperforming a prior 29-model ensemble with 3.6× fewer members. Four of the eight members are innovation variants (MoE, JK, Physics, DefType) whose individual MAEs are statistically indistinguishable from the baseline, demonstrating that the primary value of physically informed innovations lies in error decorrelation.

4. **Quantified generalization boundaries.** The graduated three-tier OOD assessment reveals graceful degradation: intra-family leave-one-host-out increases MAE by only 1.5× (0.540 eV), far below the naïve-baseline collapse (4.9×). Compositional block-out yields comparable degradation (0.534 eV), confirming robust matrix-completion generalization.

5. **Calibrated uncertainty for deployment.** Temperature-scaled ensemble uncertainty achieves 93.4% coverage at the 90% nominal level (ECE = 0.037), with a Spearman correlation of 0.577 between predicted σ and absolute error, enabling practitioners to identify unreliable predictions before deployment.

Together, these results establish DART as a reliable and efficient surrogate for accelerating defect engineering in 2D materials—from ranking candidate dopants for catalytic activation and single-photon emission to flagging deep-trap species detrimental to electronic transport—with clearly delineated applicability domains and element-resolved confidence estimates. The architectural innovations demonstrated here, particularly the gated pooling and environment enrichment modules, are transferable to other defect-property prediction tasks in both 2D and 3D settings. The code, trained models, and evaluation protocols are publicly available to support reproducibility and community benchmarking.

DATA AND CODE AVAILABILITY

The source IMP2D data are available through the repository associated with Ref. [1]. The exact sample audit, immutable split definitions, resolved experiment configurations, analysis scripts, and machine-readable result bundles for this study are available at `https://github.com/chimeraHHH/2d-defect-dast`. A release-specific commit and archival identifier will be added before publication.

ACKNOWLEDGMENTS

This work was supported by the National College Students Innovation and Entrepreneurship Training Program (Key Support Area Project). Computations were performed on an NVIDIA L40S cluster at Wuhan University.

[1] J. Davidsson, F. Bertoldo, K. S. Thygesen, and R. Armiento, Absorption versus adsorption: high-throughput computation of impurities in 2D materials, npj 2D Materials and Applications **7**, 26 (2023).
[2] K. Choudhary and B. DeCost, Atomistic line graph neural network for improved materials property predictions, npj Computational Materials **7**, 185 (2021).
[3] K. T. Schütt, H. E. Sauceda, P.-J. Kindermans, A. Tkatchenko, and K.-R. Müller, SchNet – a deep learning architecture for molecules and materials, The Journal of Chemical Physics **148**, 241722 (2018).
[4] A. Kesorn, R. Hunkao, C. Na Talang, C. Cholsuk, A. Sinsarp, T. Vogl, S. Suwanna, and S. Yuma, Formation energy prediction of neutral single-atom impurities in 2D materials using tree-based machine learning, Machine Learning: Science and Technology **5**, 035039 (2024).
[5] A. El Alouani, M. Al Khalfioui, A. Michon, S. Vézian, P. Boucaud, and M. T. Dau, Defect formation energy of impurities in 2D materials: How does data engineering shape machine learning model selection?, Materials Today Physics **61**, 102006 (2026).
[6] J. Park, J. Kim, Y. Kang, I.-H. Lee, and S. Han, DefiNet: Equivariant graph neural network for predicting defect formation energies in crystalline materials, Chemistry of Materials **37**, 694 (2025).
[7] Z. Fang and Q. Yan, Leveraging persistent homology features for accurate defect formation energy predictions via graph neural networks, Chemistry of Materials **37**, 1531 (2025).
[8] S. Kiyohara, C. Shibui, S. Bae, and Y. Kumagai, Machine-learning prediction of charged-defect formation energies from crystal structures, Physical Review Letters **135**, 246101 (2025).
[9] S. S. Omee, N. Fu, R. Dong, M. Hu, and J. Hu, Structure-based out-of-distribution (OOD) materials property prediction: a benchmark study, npj Computational Materials **10**, 144 (2024).
[10] K. Li, A. N. Rubungo, X. Lei, D. Persaud, K. Choudhary, B. DeCost, A. B. Dieng, and J. Hattrick-Simpers, Probing out-of-distribution generalization in machine learning for materials, Communications Materials **6**, 9 (2025).
[11] L. Jin, Z. Du, L. Shu, Y. Cen, Y. Xu, Y. Mei, and H. Zhang, Transformer-generated atomic embeddings to enhance prediction accuracy of crystal properties with machine learning, Nature Communications **16**, 1210 (2025).
[12] P. Izmailov, D. Podoprikhin, T. Garipov, D. Vetrov, and A. G. Wilson, Averaging weights leads to wider optima and better generalization, Proceedings of the Conference on Uncertainty in Artificial Intelligence (UAI) (2018).
[13] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, LightGBM: A highly efficient gradient boosting decision tree, in *Advances in Neural Information Processing Systems*, Vol. 30 (2017).
[14] B. Lakshminarayanan, A. Pritzel, and C. Blundell, Simple and scalable predictive uncertainty estimation using deep ensembles, in *Advances in Neural Information Processing Systems*, Vol. 30 (2017).
[15] A. N. Angelopoulos and S. Bates, Conformal prediction: A gentle introduction, Foundations and Trends in Machine Learning **16**, 494 (2023).
[16] R. El-Yaniv and Y. Wiener, On the foundations of noise-free selective classification, Journal of Machine Learning Research **11**, 1605 (2010).
[17] T. Taniai, R. Igarashi, Y. Suzuki, N. Chiba, K. Saito, Y. Ushiku, and K. Ono, Crystalformer: Infinitely connected attention for periodic structure encoding, in *International Conference on Learning Representations (ICLR)* (2024).